

Circuit Simulation Lab: 2

Binary Adders

The purpose of this experiment is to introduce the design of simple combinational circuits, in this case half adders, and full adders

Software tools

ICARUS & GTKWAVE

Task :

Describe the following basic adders in Verilog HDL and capture the Waveforms

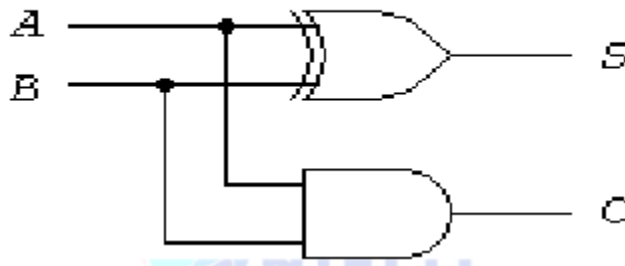
Dataflow modeling

Behavior modeling

Structural modeling

Half adder

Logic Diagram



Dataflow modeling

Code:

```
module halfadder_DF(  
input a, b ,  
output sum , carry );
```

```
assign sum = a ^ b ;  
assign carry = a & b ;  
endmodule
```

Testbench

```
module halfadder_tb;  
reg a , b ;  
wire sum , carry ;
```

```
halfadder_DF d1 (.a(a),.b(b),.sum(sum),.carry(carry));
```

```
initial begin
```

```
$dumpfile ("half_df.vcd");
```

```
$dumpvars(0,halfadder_tb);
```

```
end
```

```
initial begin
```

```
repeat (20) begin
```

```
a = $random%2 ;
```

```
b = $random % 2;
```

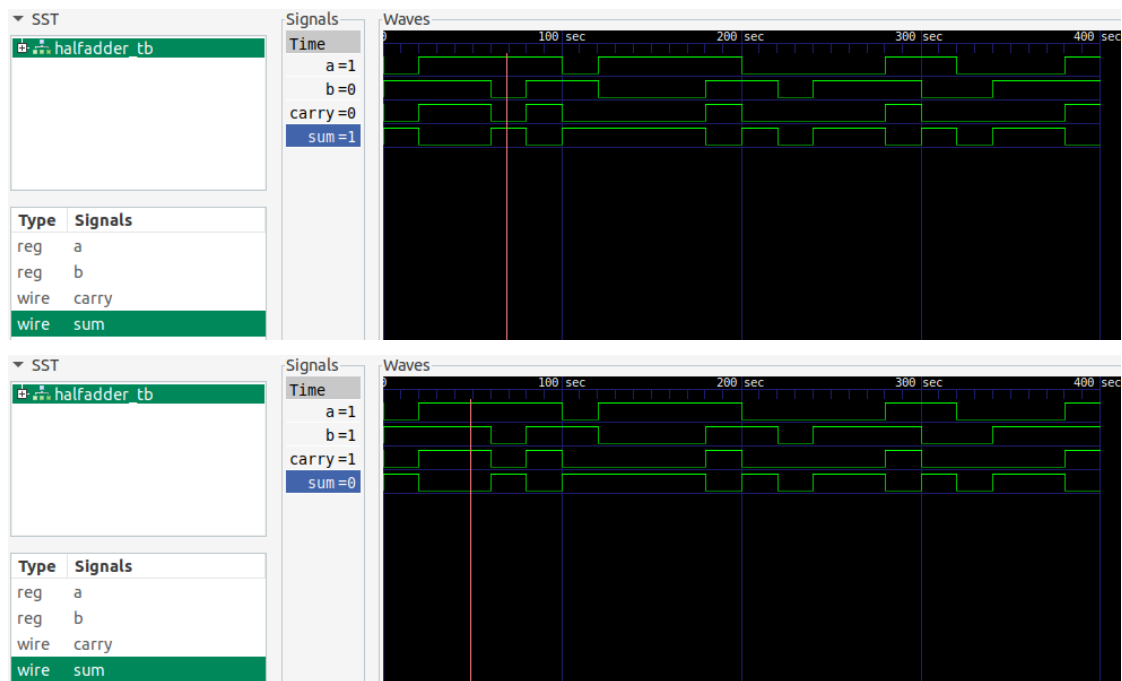
```
$display (" time = %t , a = %d , b = %d , sum = %d , carry = %d ",$time , a , b , sum , carry );
```

```
#20;
```

```
end
```

```
end
```

```
endmodule
```



```
VCD info: dumpfile half_df.vcd opened for output.
time = 0 , a = 0 , b = 1 , sum = x , carry = x
time = 20 , a = 1 , b = 1 , sum = 1 , carry = 0
time = 40 , a = 1 , b = 1 , sum = 0 , carry = 1
time = 60 , a = 1 , b = 0 , sum = 0 , carry = 1
time = 80 , a = 1 , b = 1 , sum = 1 , carry = 0
time = 100 , a = 0 , b = 1 , sum = 0 , carry = 1
time = 120 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 140 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 160 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 180 , a = 1 , b = 1 , sum = 1 , carry = 0
time = 200 , a = 0 , b = 1 , sum = 0 , carry = 1
time = 220 , a = 0 , b = 0 , sum = 1 , carry = 0
time = 240 , a = 0 , b = 1 , sum = 0 , carry = 0
time = 260 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 280 , a = 1 , b = 1 , sum = 1 , carry = 0
time = 300 , a = 1 , b = 0 , sum = 0 , carry = 1
time = 320 , a = 0 , b = 0 , sum = 1 , carry = 0
time = 340 , a = 0 , b = 1 , sum = 0 , carry = 0
time = 360 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 380 , a = 1 , b = 1 , sum = 1 , carry = 0
```

Behavior modeling

Code:

```
module halfadder_BM (  
input a , b ,  
output reg sum ,  
output reg carry );
```

```
always @(*) begin  
sum = a ^ b ;  
carry = a & b ;  
end  
endmodule
```

Testbench

```
module halfadder_tb;  
reg a , b ;  
wire sum , carry ;
```

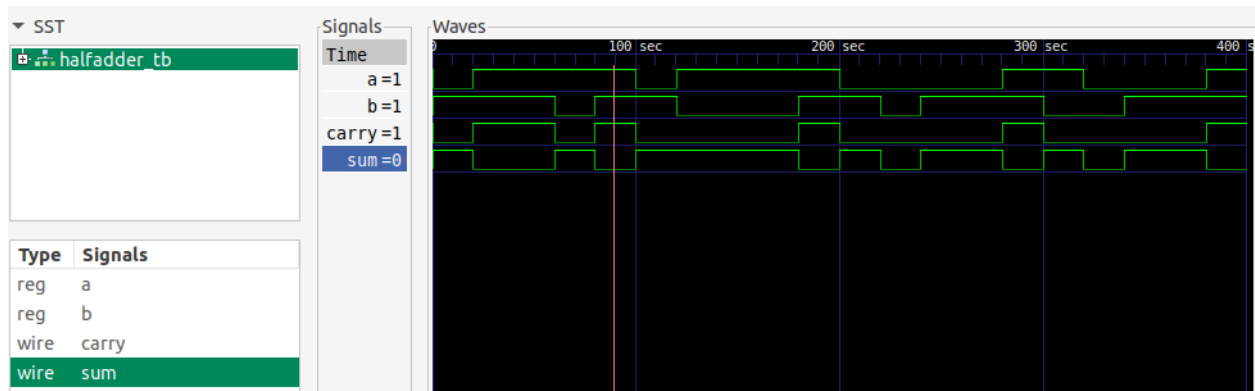
```
halfadder_BM d1 (.a(a),.b(b),.sum(sum),.carry(carry));
```

```
initial begin  
$dumpfile ("half_BM.vcd");  
$dumpvars(0,halfadder_tb);  
end
```

```
initial begin  
repeat (20) begin  
a = $random%2 ;  
b = $random % 2;  
$display (" time = %t , a = %d , b = %d , sum = %d , carry = %d ",$time , a, b , sum , carry );  
#20;  
end
```

```
end  
endmodule
```

Outputs



```

time = 20 , a = 1 , b = 1 , sum = 1 , carry = 0
time = 40 , a = 1 , b = 1 , sum = 0 , carry = 1
time = 60 , a = 1 , b = 0 , sum = 0 , carry = 1
time = 80 , a = 1 , b = 1 , sum = 1 , carry = 0
time = 100 , a = 0 , b = 1 , sum = 0 , carry = 1
time = 120 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 140 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 160 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 180 , a = 1 , b = 1 , sum = 1 , carry = 0
time = 200 , a = 0 , b = 1 , sum = 0 , carry = 1
time = 220 , a = 0 , b = 0 , sum = 1 , carry = 0
time = 240 , a = 0 , b = 1 , sum = 0 , carry = 0
time = 260 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 280 , a = 1 , b = 1 , sum = 1 , carry = 0
time = 300 , a = 1 , b = 0 , sum = 0 , carry = 1
time = 320 , a = 0 , b = 0 , sum = 1 , carry = 0
time = 340 , a = 0 , b = 1 , sum = 0 , carry = 0
time = 360 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 380 , a = 1 , b = 1 , sum = 1 , carry = 0

```

Structural modeling

Codes

```

module and1(
input in1 , in2,
output out1);
assign out1 = in1 & in2 ;
endmodule

```

```

module xor1(
input in1 , in2 ,
output out1);

```

```

assign out1 = in1 ^ in2 ;
endmodule

```

```

module halfadder_ST(
input a , b ,
output sum , carry );

and1 a1 (.in1(a),.in2(b),.out1(carry));
xor1 x1 (.in1(a),.in2(b),.out1(sum));
endmodule

```

```

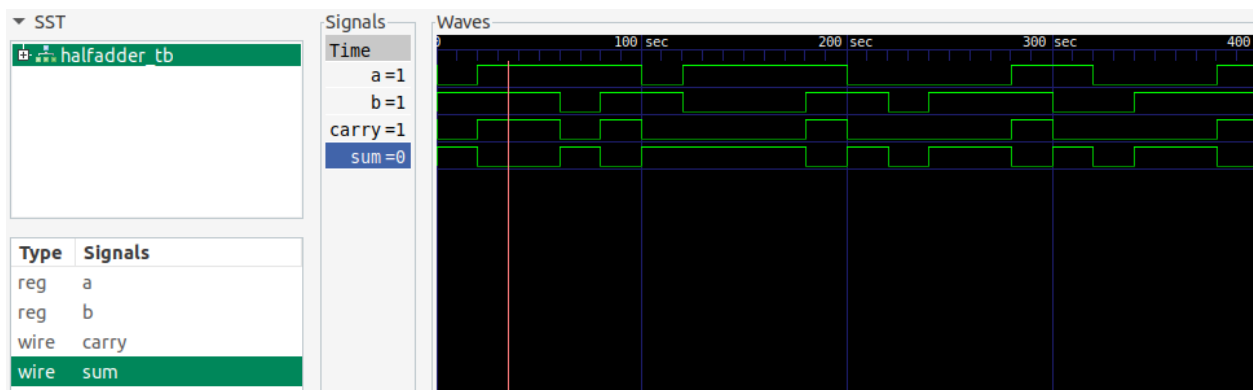
module halfadder_tb;
reg a , b ;
wire sum , carry ;

halfadder_ST d1 (.a(a),.b(b),.sum(sum),.carry(carry));

initial begin
$dumpfile ("half_BM.vcd");
$dumppvars(0,halfadder_tb);
end

initial begin
repeat (20) begin
a = $random%2 ;
b = $random % 2;
#20;
$display (" time = %t , a = %d , b = %d , sum = %d , carry = %d ",$time , a , b , sum , carry );
end
end
endmodule
Outputs

```

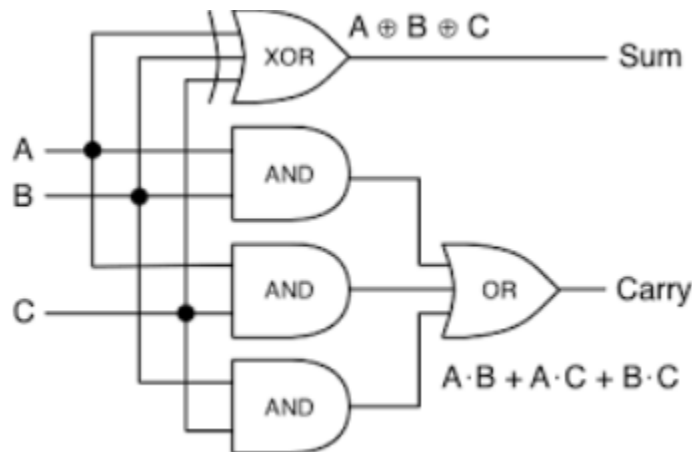


```

VCD info: dumpfile half_BM.vcd opened for output.
time = 20 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 40 , a = 1 , b = 1 , sum = 0 , carry = 1
time = 60 , a = 1 , b = 1 , sum = 0 , carry = 1
time = 80 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 100 , a = 1 , b = 1 , sum = 0 , carry = 1
time = 120 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 140 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 160 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 180 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 200 , a = 1 , b = 1 , sum = 0 , carry = 1
time = 220 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 240 , a = 0 , b = 0 , sum = 0 , carry = 0
time = 260 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 280 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 300 , a = 1 , b = 1 , sum = 0 , carry = 1
time = 320 , a = 1 , b = 0 , sum = 1 , carry = 0
time = 340 , a = 0 , b = 0 , sum = 0 , carry = 0
time = 360 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 380 , a = 0 , b = 1 , sum = 1 , carry = 0
time = 400 , a = 1 , b = 1 , sum = 0 , carry = 1

```

Full adder



Full adder

Dataflow modeling

Code:

```
module fulladder_DF(  
input a , b , cin,  
output carry , sum );
```

```
    assign sum = a ^ b ^ cin;  
    assign carry = (a & b ) | (cin & a ) | (cin & b );
```

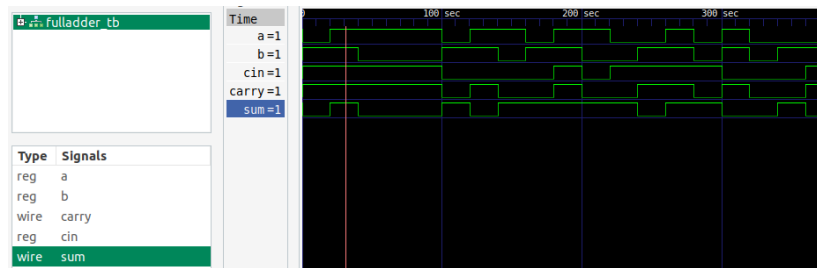
```
endmodule
```

Tb

```
module fulladder_tb;  
reg a , b , cin ;  
output sum , carry ;
```

```
    fulladder_DF d1 (.a(a),.b(b),.cin(cin),.sum(sum),.carry(carry));
```

```
initial begin  
    $dumpfile ("fulladder.vcd");  
    $dumpvars(0,fulladder_tb);  
    repeat (20) begin  
        a = $random %2;  
        b = $random %2;  
        $display (" time = %t , a = %d , b = %d , cin = %d ,sum = %d , carry = %d ",$time , a , b ,cin, sum  
        , carry );  
        #20;  
    end  
end  
endmodule
```



```
VCD info: dumpfile fulladder.vcd opened for output.
time = 0 , a = 0 , b = 1 , cin = 1 ,sum = x , carry = x
time = 20 , a = 1 , b = 1 , cin = 1 ,sum = 0 , carry = 1
time = 40 , a = 1 , b = 0 , cin = 1 ,sum = 1 , carry = 1
time = 60 , a = 1 , b = 0 , cin = 1 ,sum = 0 , carry = 1
time = 80 , a = 1 , b = 0 , cin = 1 ,sum = 0 , carry = 1
time = 100 , a = 0 , b = 1 , cin = 0 ,sum = 0 , carry = 1
time = 120 , a = 1 , b = 1 , cin = 0 ,sum = 1 , carry = 0
time = 140 , a = 1 , b = 0 , cin = 0 ,sum = 0 , carry = 1
time = 160 , a = 0 , b = 1 , cin = 0 ,sum = 1 , carry = 0
time = 180 , a = 1 , b = 1 , cin = 1 ,sum = 1 , carry = 0
time = 200 , a = 1 , b = 0 , cin = 0 ,sum = 1 , carry = 1
time = 220 , a = 0 , b = 0 , cin = 1 ,sum = 1 , carry = 0
time = 240 , a = 0 , b = 1 , cin = 1 ,sum = 1 , carry = 0
time = 260 , a = 1 , b = 1 , cin = 1 ,sum = 0 , carry = 1
time = 280 , a = 0 , b = 0 , cin = 1 ,sum = 1 , carry = 1
time = 300 , a = 1 , b = 1 , cin = 0 ,sum = 1 , carry = 0
time = 320 , a = 0 , b = 0 , cin = 0 ,sum = 0 , carry = 1
time = 340 , a = 0 , b = 1 , cin = 0 ,sum = 0 , carry = 0
time = 360 , a = 0 , b = 1 , cin = 1 ,sum = 1 , carry = 0
time = 380 , a = 0 , b = 0 , cin = 0 ,sum = 0 , carry = 1
```

Behavior modeling

```
module fulladder_BM(
```

```
input a , b , cin ,
```

```
output reg sum ,
```

```
output reg carry );
```

```
always @(*) begin
```

```
sum = a ^ b ^ cin ;
```

```
carry = (a & cin ) | (a & b ) | ( cin & b );
```

```
end
```

```
endmodule
```

```
module fulladder_tb;
```

```
reg a , b , cin ;
```

```
output sum , carry ;
```

```
fulladder_BM d1 (.a(a),.b(b),.cin(cin),.sum(sum),.carry(carry));
```

```
initial begin
```

```
$dumpfile ("fulladder.vcd");
```

```
$dumpvars(0,fulladder_tb);
```

```
repeat (20) begin
```

```
a = $random %2;
```

```
b = $random %2;
```

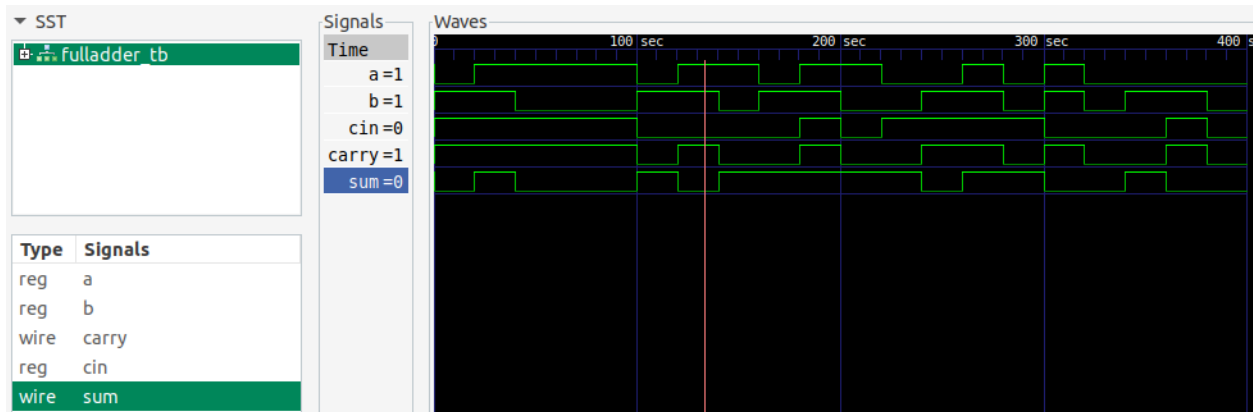
```
cin = $random %2;
```

```
$display (" time = %t , a = %d , b = %d , cin = %d ,sum = %d , carry = %d ",$time , a , b ,cin, sum , carry );
```

```

#20;
end
end
endmodule

```



Structural modeling

```

module or1 (
input a , b ,
output out1);
assign out1 = a | b ;
endmodule

```

```

module halfadder_ST(
input a , b ,
output sum , carry );

```

```

assign sum = a ^ b ;
assign carry = a & b ;
endmodule

```

```

module fulladder_ST(
input a , b , cin ,
output sum , carry );

```

```

wire sum1 ;
wire carry1 ;
wire carry2 ;
halfadder_ST h1 (.a(a),.b(b),.sum(sum1),.carry(carry1));
halfadder_ST h2 (.a(sum1),.b(cin),.sum(sum),.carry(carry2));
or1 o1 (.a(carry1),.b(carry2),.out1(carry));
endmodule

```

